1. This question covers several topics and all parts should be attempted.

a) Consider an overhead line used for a single-phase AC supply. The line has a series impedance of $j30\ \Omega$, that is, it is purely inductive.

i) Calculate the real and reactive powers at the sending end if the sending end voltage is 220 kV and the receiving end voltage is 215 kV lagging the sending end by 6.0°. [6]

[Use of standard formula]

$$P = \frac{V_R V_S}{X} sin(\delta) = \frac{220 \times 10^3 \times 215 \times 10^3}{30} \sin(6.0°) = 164.8\ MW$$

$$Q_S \approx \frac{{V_S}^2}{X} - \frac{V_R V_S}{X} \cos(\delta)$$
$$= \frac{220 \times 10^3 \times 220 \times 10^3}{30} - \frac{220 \times 10^3 \times 215 \times 10^3}{30} \cos(6.0°)$$
$$= 45.3\ Mvar$$

[3 marks for each correct answer.]

ii) Calculate the maximum real power this line can transfer when operated at 220 kV at both ends if the only limit on power transfer is the angle criteria. [4]

[Use of standard formula]

Maximum power when angle is 90° and the receiving end voltage is raised to 220 kV.

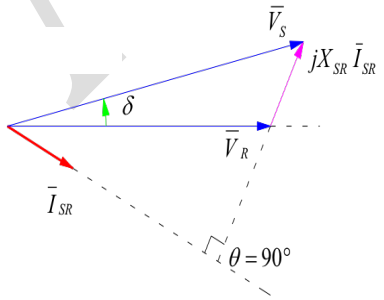
$$P = \frac{V_R V_S}{X} \sin(90°) = \frac{220 \times 10^3 \times 220 \times 10^3}{30} = 4{,}500\ MW$$

[2 mark for using 90°; 2 mark for correct answer.]

iii) The line has a maximum current carrying capability of 3,500 A. For operation of this line under this limit, and with both ends at 220 kV, calculate the load angle at which the line would operate and the power transfer this creates. [6]

[Use of standard method]

Angle between sending and receiving ends now set my maximum length of the voltage phasor across the line impedance. The cosine rule for the triangle of the phasor diagram can be used.




$$(I_S X_{SR})^2 = V_S^2 + V_R^2 - 2V_S V_R cos(\delta)$$

$$cos(\delta) = 1 - \frac{(I_S X_{SR})^2}{2V^2} = 1 - \frac{(1{,}500 \times 30)^2}{2 \times 220{,}000^2} = 0.8861$$

Answers

$$\delta = 27.61°$$

[4 marks]

$$\mathrm{P} = \frac{V_R V_S}{X} sin(\delta) = \frac{220 \times 10^3 \times 220 \times 10^3}{30} \sin(11.74°) = 747.7\ MW$$

[2 marks]

b) Consider a three-phase AC supply with a line voltage of 11 kV.

i) Three impedances of (6.0 – $j$ 3.5) Ω are formed into a star connection and connected to the supply. Calculate the phase current and line currents that flow. Provide magnitudes of both currents and angle of a phase current relative to a phase voltage at 0°. [4]

[Standard calculation aimed at testing basic ability to work with three-phase systems. Similar to the question in 2020 that was for a delta connection]

First find phase voltage magnitude

$$V_P = \frac{V_L}{\sqrt{3}} = 6.35\ kV$$

Now find phase current

$$I_P = \frac{V_P}{Z} = \frac{6{,}350}{6.0 - j3.5} = 914.2\angle 30.25°\ \mathrm{A}$$

[1 mark for method; 1 mark for magnitude and 1 mark for angle including correct (positive) sign.]

Line current and phase current are equivalent in star connection. [1 mark]

$$|I_L| = |I_P| = 914.2\ A$$

ii) Calculate the real and reactive powers drawn by the star-connected impedances. [4]

[Standard calculation aimed at testing basic ability to work with real and reactive power]

Take "power factor" angle as the angle of the impedance from part (i), that is -30.25°, or use negation of angle of current.

$$P = 3\,V_P I_P \cos(\theta) = 3 \times 6{,}350 \times 914.2\,\cos(-30.25°) = 15.05\ MW$$

$$Q = 3\,V_P I_P \sin(\theta) = 3 \times 6{,}350 \times 914.2\,\sin(-30.25°) = -8.77\ Mvar$$

[1 mark for formulas; 2 marks for accurate answers; 1 mark for correct units for both answers.]

iii) One of the three impedances is found to be (5.5 – j 3.0) Ω (the other two impedance remain at (6.0 – $j$ 3.5) Ω. Calculate the magnitude of the current that flows in the neutral in this situation. [4]

[Relatively simple case of unbalanced load.]

In star connection with a neutral wire, the phase voltages will be held as a balanced set at 6.35 kV and so to phase currents remain unchanged. One phase current changes and the change in current will flow in the neutral because the original portion of the current sums to zero with the other two phase currents.

Answers

Original phase current was $I_P = \frac{V_P}{Z} = \frac{6{,}350}{6.0-j3.5} = 914.2\angle 30.25° = 789.7 - j460.7\ A$

New phase current is $I'_P = \frac{V_P}{Z} = \frac{6{,}350}{5.5-j3.0} = 1{,}014\angle 28.61° = 889.3 - j485.4\ A$

Neutral current is $I'_P - I_P = 100.2 - j24.7\ A$

Magnitude of the neutral current is 103.2 A

[2 mark for method; 2 marks for accurate answers.]

c) Consider an induction machine.

i) Explain what determines the rate of change of flux linkage of the rotor bars of an induction machine. [4]

[Interpretation of notes]

- The magnitude of the magnetic flux linking the rotor circuit (the flux crossing the air-gap plus rotor leakage flux) [1 mark]
- The number of turns of the rotor circuit [1 mark]
- The rotational speed difference between the stator flux (the synchronous speed) and the physical speed of rotation of the rotor. [2 marks]

ii) Explain what power loss the magnetising resistance represents. [4]

[Interpretation of notes]

This represents the power losses caused by hysteresis in the core material (steel) of the stator and rotor plus power losses caused by eddy currents in the core material caused by the changing flux.

[2 marks for each item]

iii) Explain what is meant by flux leakage, how it is represented in an equivalent circuit and why it does not directly impact the efficiency of the machine. [4]

[Interpretation of notes]
Flux leakage is flux created by a winding, the rotor winding for instance, that does not link with the other winding, the stator winding in this example, and therefore does not contribute to the action of the induction machine.

[2 marks

It is modelled in the equivalent circuit as additional inductances in series with the stator and rotor windings either side of the magnetising inductance (which models the flux linking both windings).

[1 mark]

The fluxes in the leakage paths create voltage drops in the windings that are in quadrature to the current, as they are for any inductance, and do not dissipate energy. They create (small amounts of) energy storage in the leakage field that rises and falls over the AC cycle.

[1 mark]

ANSWERS

2. This question concerns **buck** switch-mode power supplies (SMPS)

a) i) The voltage transfer ratio of buck SMPS is often quoted as $\frac{V_O}{V_I} = \delta$ where $\delta$ is the duty-cycle of the switch. Explain the basis on which this equation is derived (you need not reproduce the derivation itself) and through that explain why calculation of output voltage using this expression is not completely accurate in practice. [5]

[Interpretation of information in the notes.]

The expression is derived using the principle that in periodic steady-state, the change in inductor current over a switching cycle will be zero. The changes in the on and off periods are found from the voltages imposed across the inductor and their sum equated to zero. [2 marks]

This version of the expression applied for continuous conduction mode and a different expression applies for discontinuous conduction mode. [1 mark]

The reasons this equation is inaccurate in practice is that it neglects:

- The on-state voltage drop of the MOSFET
- The conduction voltage of the diode
- Voltage drop across the resistance of the inductors

[2 marks of all three reasons, 1 mark for two]

ii) Derive an expression for the output voltage of a Buck SMPS in continuous conduction that accounts for voltage drops within the circuit. [5]

[Extension of standard derivation in which neglected terms are retained. Principle was shown in examples class.]

For the on-state [1 mark]

$$\Delta i_L^{on} = \frac{V_I - I_L R_L - I_L R_{DS} - V_O}{L}\delta T$$

For the off state [1 mark]

$$\Delta i_L^{off} = \frac{-I_L R_L - V_D - V_O}{L}(1-\delta)T$$

Adding and equating to zero gives

$$0 = V_I\delta - I_L R_L - I_L R_{DS}\delta - V_O\delta - I_L R_L(1-\delta) - V_D(1-\delta) - V_O(1-\delta)$$

Gathering terms gives

$$0 = V_I\delta - I_L R_L - I_L R_{DS}\delta - V_O - V_D(1-\delta)$$

Re-arranging gives

$$V_O = V_I\delta - I_L R_L - I_L R_{DS}\delta - V_D(1-\delta)$$

[3 marks for final expression]

In which the first term is the ideal expression and is followed by the resistance of the inductor, which is always in circuit; the MOSFET channel resistance, present only during the on-time, and the diode voltage drop, present only during the off-time.

Answers

b) A **buck** SMPS is to be designed to convert 24.0 V to 5.0 V with output currents of up to 8.0 A.

The on-state resistance of the chosen MOSFET is 0.14 Ω. The turn-on and turn-off energy losses of the MOSFET are 1.5 µJ and 1 µJ when operated at 1 A and **scale linearly with current**. The MOSFET is mounted on a heat-sink which gives a total thermal impedance between junction and air of $R_{ThJA} = 12.5\ °C/W$.

The SMPS is required to work in ambient temperatures of up to 45°C and the junction temperature is to be no greater than 105°C.

i) Calculate the maximum switching frequency at which the MOSFET can be operated in this design. You may assume the SMPS will be in continuous conduction mode when operating at maximum current. [6]

[Calculation using standard method.]

First calculate the maximum allowable power loss for the thermal conditions [2 marks]

$$P_{M\ Loss}^{Max} = \frac{\theta_J^{Max} - \theta_J}{R_{Th(JA)}} = \frac{95 - 45}{12.5} = 4.0\ W$$

Next find conduction more loss of MOSFET [2 marks]

$$P_{M\ Cond} = R_{DS(On)} I_L^2 \delta = 0.14 \times 8^2 \times \frac{5.0}{24.0} = 1.867\ W$$

Switching loss is therefore 4.0-1.867 = 2.133 W. Now find frequency given the energy loss. [2 marks]

$$f = \frac{P_{M\ Loss}^{Max} - P_{M\ Cond}}{E_{On} + E_{On}} = \frac{2.133}{8 \times (1.5 \times 10^{-6} + 1 \times 10^{-6})} = 106.7\ kHz$$

ii) The capacitor suggested for the design has a capacitance of $C = 750\ \mu F$ and effective series resistance of $R_{ESR} = 35\ m\Omega$. The output voltage ripple is not to exceed $\Delta v_O = 10\ mV$. Express the total output voltage ripple as the simple sum of terms due to the capacitance and effective series resistance and use this to find the maximum inductor current ripple that can be allowed. [4]

[Expression needed is combination of two terms from notes.]

$$\Delta v_O = \Delta v_C + \Delta v_{ESR} = \frac{\Delta i_L}{8fC} + R_{ESR} \Delta i_L$$

[2 marks]

$$\Delta i_L = \frac{\Delta v_O}{\frac{1}{8fC} + R_{ESR}} = \frac{0.01}{\frac{1}{8 \times 106.7 \times 10^3 \times 750 \times 10^{-6}} + 0.035} = 0.274\ A$$

[2 marks for accurate answer]

iii) Calculate the value of inductor that should be chosen. [4]

[Standard Calculation]

Answers

$$L = \frac{V_I - V_O}{\Delta i_L}\frac{\delta}{f} = \frac{24-5}{0.274}\frac{\frac{5}{24}}{106.7\times10^3} = 135\ \mu H$$

iv) Assume the inductor has a resistance of $R_L = 0.05\ \Omega$ and the diode has a forward voltage drop of $V_D = 0.8\ V$. Calculate the efficiency of the SMPS at full load. [4]

[Variation of standard calculation]

Mosfet power losses were configured in part (i) to be 4.0 W

Inductor loss is $P_{RL} = R_L I_L^2 = 0.03 \times 8^2 = 1.92\ W$

[1 mark]

Diode loss is $P_D = V_D I_L(1-\delta) = 0.8 \times 8 \times \left(1 - \frac{5}{24}\right) = 5.07\ W$

[1 mark]

Total power loss is 4.0 + 1.92 + 5.07 = 10.98 W

Efficiency is $\eta = \frac{P_O}{P_O + P_{Loss}} = \frac{5\times8}{5\times8+10.98} = 78.4\%$

[2 marks for accurate answer]

v) It is suggested that to raise the efficiency, synchronous rectification could be used with a second MOSFET of the same characteristics as the main MOSFET. How effective would that be? You may assume that the second, rectifier, MOSFET incurs conduction power loss but no switching power loss. [2]

[New interpretation of facts.]

The on-state resistance of the MOSFET is 0.14 Ω which at 8 A give a voltage drop of 1.12 V which is larger than the diode. The rectifier MOSFET will not conduct because the diode is a better path and the efficiency will not be improved. The MOSFET would need a lower resistance for synchronous rectification to be beneficial.

[2 marks for correct discussion of the issues].

Answers

3. An induction machine which is a 3-phase, 2 pole-pair, star-connected machine is connected to a 50 Hz supply with a phase-voltage of 230 V. The parameters of the equivalent circuit, referred to the stator, are:

| | |
|---|---|
| Stator resistance, | $R_S = 0.46\ \Omega$; |
| Stator leakage reactance, | $X_S = 0.51\ \Omega$; |
| Magnetising resistance, | $R_M = 500\ \Omega$; |
| Magnetising reactance, | $X_M = 60\ \Omega$; |
| Referred rotor leakage reactance, | $X'_R = 0.51\ \Omega$; |
| Referred rotor resistance, | $R'_R = 0.60\ \Omega$. |

The machine is operated with a slip of $s = 0.02$.

i) Give an expression for the total input impedance of the machine and calculate its value. [5]

[2 marks for correct formulation of impedance, including correct parallel terms and inclusion of $s$ with those errors leading to loss of all marks.]

Total impedance: $Z_T = R_S + jX_S + \left(R_M // jX_M // \left(\frac{R'_R}{s} + jX'_R\right)\right)$

[3 marks for accurate computation.]

$$Z_T = 23.33 + \text{j}11.67 = 26.08\angle 26.60°\ \Omega$$

ii) Calculate the magnitude of the rotor current. [5]

[Answer from previous part will need to be carried forward; full marks can be obtained here if method is sound.]

First, find stator current from the phase voltage

$$I_S = \frac{V_S}{Z_T} = \frac{230}{26.08\angle 26.60°} = 8.82\angle -26.60°\ \text{A}$$

[For reference, rectangular form is 7.87 – j 3.94 A]

[2 marks for accurate answer.]

Then rotor current is found from the current divider rule (or otherwise)

$$I'_R = I_S \times \left(\frac{R_M // jX_M}{R_M // jX_M + \left(\frac{R'_R}{s} + jX'_R\right)}\right) = 7.48\angle -1.54°\ \text{A}$$

[1 mark for correct formulation; 2 mark for accurate answer. 1 mark lost if only rectangular form was quoted since question was explicit about asking for magnitude.]

[For reference, rectangular form is 7.48 – j 0.20 A]

iii) Calculate the electromagnetic torque developed. [5]

[Answer from previous part will need to be carried forward; full marks can be obtained here if method is sound.]

Answers

Power from standard formula $P_{EM} = 3\, I'^2_R\, R'_R \left(\frac{1-s}{s}\right) = 3 \times 7.48 \times 0.6 \times \left(\frac{1-0.02}{0.02}\right) =$ 4.93 kW

[1 mark for correct formula; 1 mark for accurate answer]

Torque found by dividing by rotor speed, which is $\omega_R = \frac{2\pi f}{P}(1-s) = 153.94\, rad/s$

$$T = \frac{P_{EM}}{\omega_R} = \frac{4{,}930}{153.94} = 32.04 \text{ Nm}$$

[1 mark for formula; 2 marks for accurate answer]

[ Alternative and more direct route via $T_{EM} = \frac{3\, I'^2_R\, R'_R}{s\, \omega_S}$ will gain full marks.]

iv) An approximate expression for funding torque, for small values of slip $s$, is $T \approx \frac{3\,|V_{Th}|^2}{\omega_S\, R'_R}\, s$ where $V_{Th}$ is the Thévenin equivalent voltage found by removing the referred rotor impedance from the circuit and taking the Thévenin equivalent voltage at that position in the circuit. Compare the torque found from this approximation to the torque found in (iii). [5]

Thévenin voltage is

$$V_{Th} = V_S \left(\frac{R_M // jX_M}{R_M // jX_M + R_s + jX_S}\right) = 227.8 + j\,1.5003 = 227.8 \angle -0.38°$$

[2 marks for correct formulation]

New torque found as

$$T \approx \frac{3\,|V_{Th}|^2}{\omega_S\, R'_R}\, s = \frac{3 \times 227.8^2}{2\pi 50 \times 0.6} \times 0.02 = 33.05\, Nm$$

[2 marks for numerical answer]

Approximation is 3.15% greater than exact answer.

[1 mark for comparison]

v) The machine is redesigned with thicker conductors in the rotor which halve the referred rotor resistance. Using the approximation in (iv) to estimate the slip at which the machine would operate to develop the same torque as in (iv). [3]

By inspection of the approximation, if resistance is halved, slip is halved for the same torque.

Or it can be found by simple rearrangement noting that the new resistance is 0.3 Ω.

$$s \approx \frac{\omega_S\, R'_R}{3\,|V_{Th}|^2}\, T = \frac{2\pi 50 \times 0.3}{3 \times 227.8^2} = 0.01$$

[3 marks]

[Alternative and acceptable answer is to use the torque found in (iii) is used instead of the torque found in (iv) and then the slip is 0.0097.]

vi) For the condition in (v), calculate the rotor current. Hint: the approximation in (iv) neglects the Thevenin impedance and that approximation can be used here. [2]

Answers

$$I_R \approx \frac{V_{Th}}{\frac{R'_R}{s}} = \frac{227.8}{0.3/0.01} = 7.59\ A$$

[2 marks]

[Full marks also given for answers that include $X'_R$ or $X_{Th}$ and are less approximate or exact.]

vii) Comment on how the power losses in the rotor resistance, stator resistance and magnetising power loss, and the efficiency of the new machine would be expected to compare to the original machine. There is no need to calculate values. [5]

The rotor has the same current in both cases but the power loss is halved because resistance has been halved.

[2 marks]

The stator current magnitude is very similar to that of the rotor current because the magnetising current is small and so the loss is unchanged.

[1 mark]

The magnetising power loss is unchanged because this is largely determined by the voltage not the current.

[1 mark]

The efficiency will be a little improved since one of the power loss terms is reduced and the output power is unchanged.

[1 mark]

Answers